1. Problem statement and our understanding

Vanguard India asks for a digital experience that teaches financial literacy to Grade 11 and 12 students, particularly from underserved communities across Hyderabad, starting with about 1,500 students and designed to scale. It must be simple, relatable, interactive and safe to explore, and cover four pillars: **plan and budget**, **save and manage**, **understand investing** and **stay financially safe**. The guardrail is to educate, never recommend investments, and to treat simulations as assumptions, not promises.

Our understanding:

- Information is already on the internet. What students lack is **experience**: money decisions only make sense when you see their consequences, and in real life those take years.
- A common belief is that a bank FD is always the safest and best option. Students rarely see how inflation reduces its real value, or how time and diversification grow long-term savings.
- Teenagers are prime targets for fake job offers, OTP calls, loan apps and "double your money" schemes, so safety must be practised, not just read about.
- The solution has to work on low-cost phones, weak internet and in local languages, reach students through NGOs, and collect no sensitive data about minors.

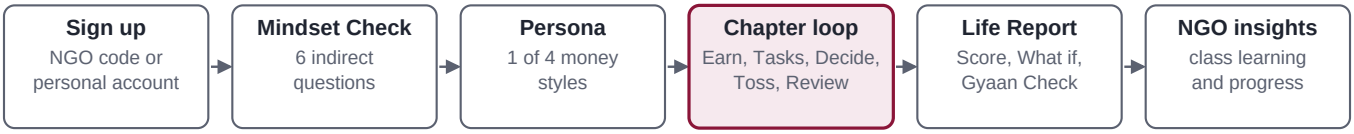
2. Idea and abstract

Our solution is a multiplayer life-simulation game. Students live a whole financial life with virtual money, from first earnings to retirement. Each **chapter is a life stage** (First Earnings, First Job, Salaried or Business path, Legacy). Every chapter gives income and a task list with due dates; students decide to spend, save, invest or lend, then time fast-forwards and a **coin toss** decides luck: market years and life events. Luck is random, but preparation decides how much it hurts or helps. A **Life Report** ends each chapter with what went well, what could be better, and a **"What if" replay** showing how one different decision would have changed the outcome.

Feature	What it does
Money Mindset Check	6 indirect, everyday questions (e.g., "Would you spend Rs 500 on a friend's gift?") place each student in one of 4 personas without asking about income or family. Repeated at the end to measure learning.
Investing at the core	Options unlock chapter by chapter: savings, RD, FD, gold, SIP in equity, debt and index funds. Every amount shows its **Real Value** after inflation, so the FD myth breaks on its own.
Friends' Room	Classmates lend to each other at interest they set, build a Trust Score (like a credit score), and bid for opportunities. A **Pool Bid** reveals how a mutual fund works.
Scams in context	Fraud offers appear as messages, calls and tempting investments, often right after a bad toss. Falling for one shows a Red Flags card; spotting the repeat earns a badge.
NGO integration	NGO admins create student codes and see learning gain, pillar mastery and scam awareness.

3. High-level approach, pipeline and architecture

Player pipeline. Every chapter repeats one simple loop, so students learn the controls once and then focus on decisions:

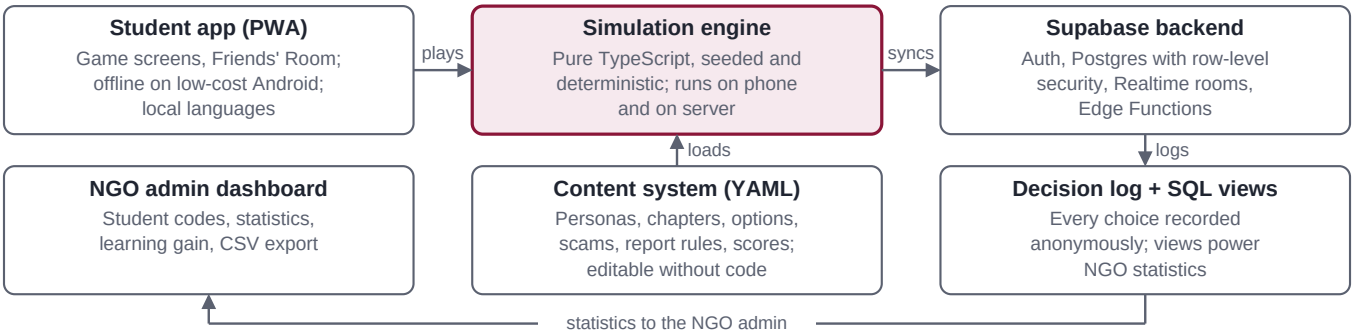


Next chapter unlocks when the Financial Health Score passes; otherwise replay with a tip.

Key design decisions

- **Seeded randomness:** every game has a seed, so the "What if" replay changes only one decision and the difference is purely due to that choice.
- **Assumptions, not promises:** every return carries an "assumed" label; market options can lose money in bad years; only fictional fund and bank names are used.
- **Privacy by design:** students use nicknames and avatars; no names, phone numbers, income or location are collected. Multiplayer uses preset emotes only, no free chat.
- **Access:** installable web app (PWA) that plays offline on low-cost phones; English and Telugu first, with Hindi supported by design.

System architecture. One deterministic engine powers both single-player and multiplayer. All game content lives in editable files, so personas, questions, chapters, returns and scams can change throughout development without code changes. Stack: Next.js PWA, TypeScript, Tailwind, Supabase and Vercel; multiplayer is server-authoritative.



Example: one student's First Job chapter (illustrative numbers)

Step	What happens	What the student learns
Earn and plan	Salary of Rs 30,000 a month. Tasks: rent, a sister's college fees by year 2, an emergency fund, an optional laptop and bike.	Needs vs. wants, due dates, goals
Decide	Puts some money in an FD, starts a small monthly SIP, keeps a safety fund in savings, skips an expensive gift.	Spreading money by goal and time
Toss	Year 2 is a market storm: the SIP dips while the FD holds. A Rs 30,000 medical bill arrives.	Risk, and why a safety net matters
Scam	Right after the bill, a call asks for an OTP to "unblock" the bank account.	Spotting red flags under pressure
Friends' Room	A classmate is short on fees and asks for a loan; the student sets 2% a month and is repaid on time.	Interest, trust and lending risk
Life Report	Score 74. Did well: safety fund covered the bill. What if: keeping the SIP from year 1 would have left about Rs 18,000 more in Real Value.	Compounding, Real Value vs. FD

4. Planned testing methodology

Level	What we test	How
Engine unit tests	Inflation and Real Value, returns per market state, SIPs, FD penalties, task fines, scams, loans and interest, auctions and pool bids, Health Score, Life Report rules	Vitest; fixed seeds give exact, repeatable results
Determinism	Same seed and same decisions always give the same result; "What if" changes only one decision	Replay tests on recorded games
Content validation	Every content file: schema, broken references, expected returns match the shown assumptions, score weights sum to 100, translation coverage	Automatic validator in CI on every change
Game balance	A balanced player usually passes; all-cash usually fails growth; all-equity with no safety net sometimes fails after bad tosses	Simulation of 1,000+ games per bot strategy
Security and privacy	Students see only their own data, admins only their NGO; codes are single use and rate limited; multiplayer actions validated on the server	Database permission tests, API tests
End-to-end	NGO signs up, creates a code, student joins, plays Chapter 2 to the Life Report, admin sees the results; multiplayer with bots	Playwright on every pull request
Device and access	Low-end Android at 360 px, slow 3G and offline play, Telugu and Hindi layouts, screen readers and contrast	Manual device testing, accessibility audit
Learning impact	Change in Mindset Check answers before vs. after, Gyaan Check accuracy, repeat-scam spotting; usability with a small student group	Playtests with students and NGO feedback

5. Expected impact and scale

- **For students and families:** confidence to budget, keep a safety net, invest for long-term goals and refuse scams, built by practice; lessons like inflation travel home.
- **For NGOs:** a ready-to-run classroom activity with discussion prompts and clear evidence of learning, without collecting personal data.
- **Measured, not assumed:** the same 6 questions before and after play show what changed for each class.
- **Built to scale:** starts with about 1,500 students; new chapters, languages and scam types are added by editing content files, and any NGO can onboard with student codes.